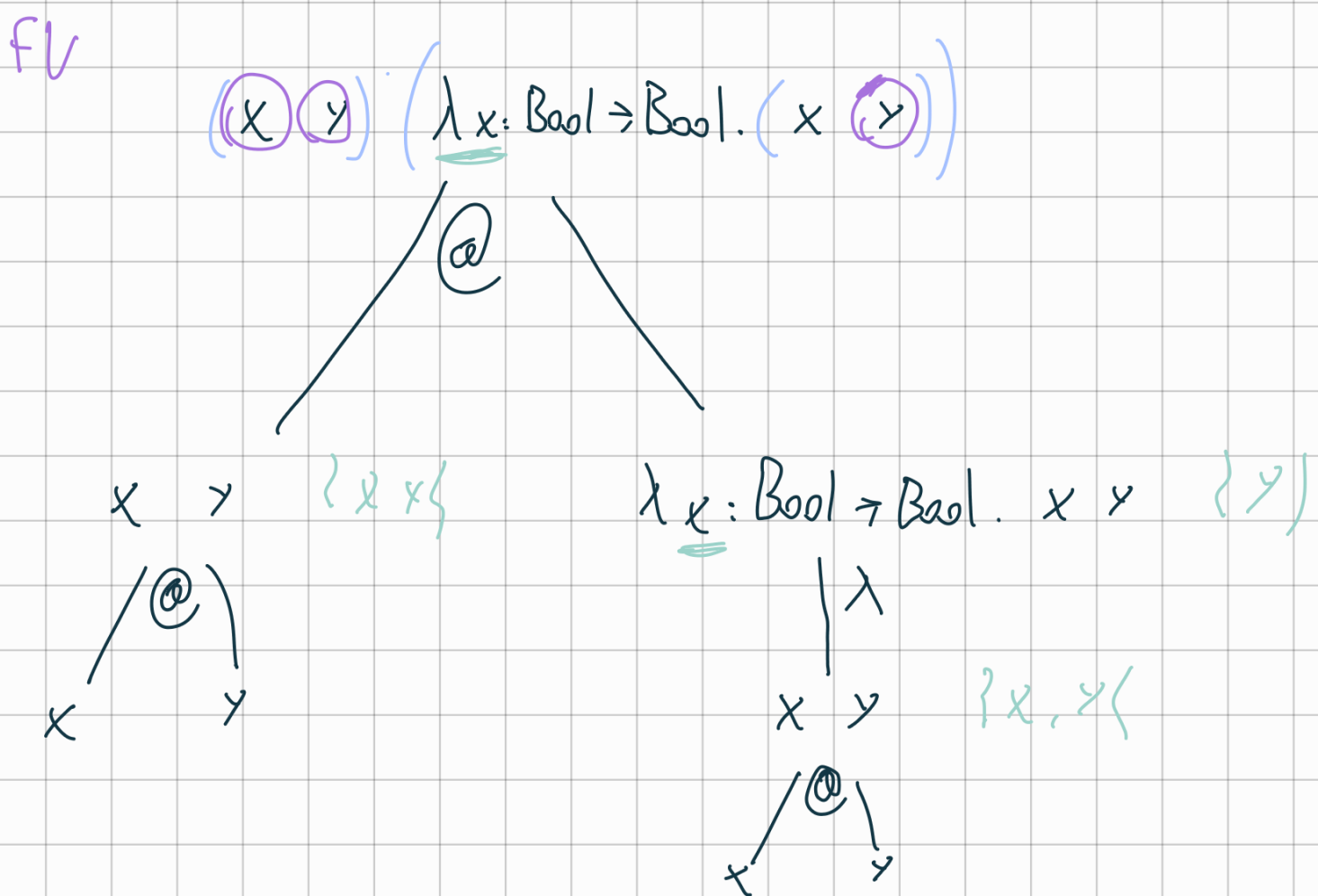
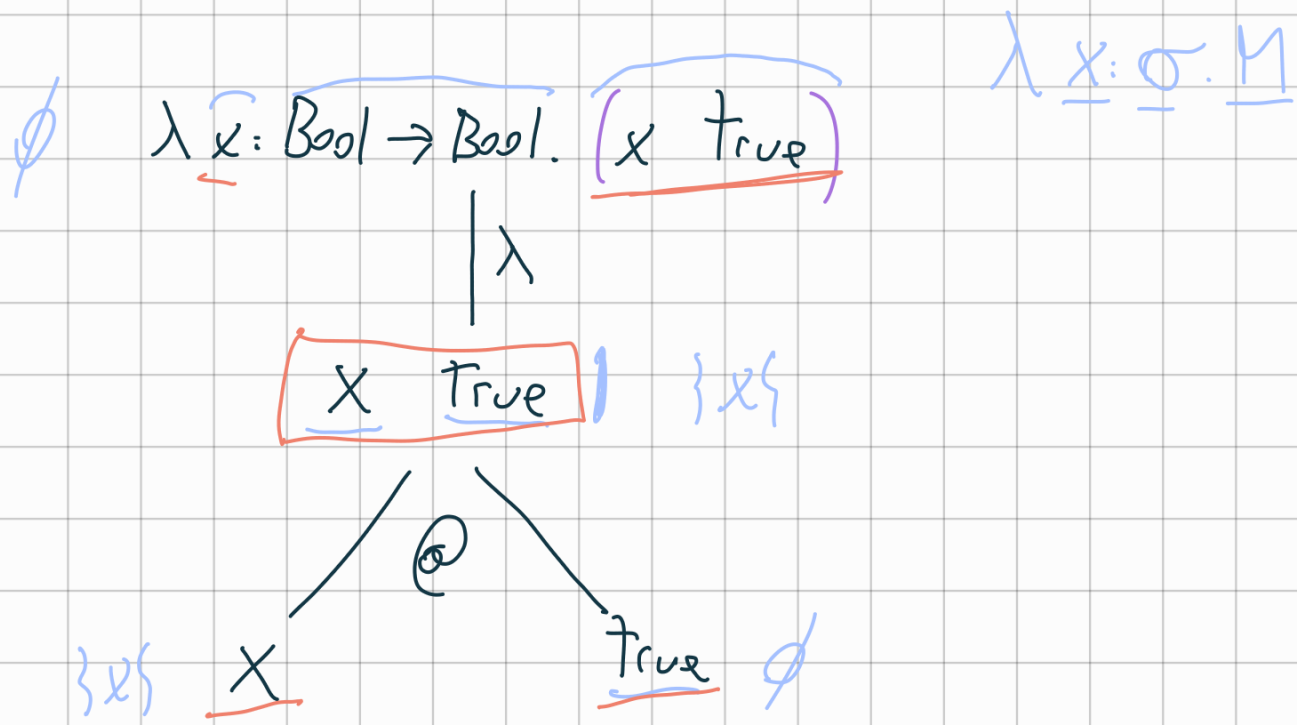
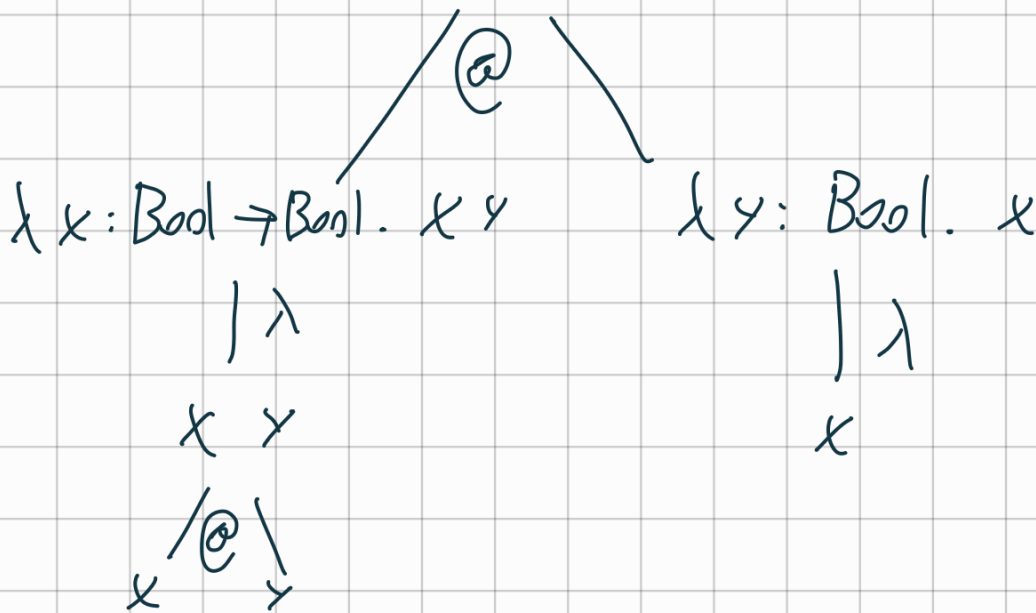


Ejercicio: ¿Cuáles de las siguientes expresiones son términos del cálculo lambda? En los casos que sí lo sean, dibujar su árbol sintáctico y marcar las ocurrencias libres de las variables.

- a) $\lambda x : \text{Bool} \rightarrow \text{Bool}. x \text{ true}$ ✓
b) $x \ y \ \lambda x : \text{Bool} \rightarrow \text{Bool}. x \ y$ ✓
c) $(\lambda x : \text{Bool} \rightarrow \text{Bool}. x \ y)(\lambda y : \text{Bool}. x)$ ✓
d) $\lambda x : \text{Bool}$ ✗
e) $\lambda x. x$ ✗
- f) $\text{if } x \text{ then } y \text{ else } \lambda z : \text{Bool}. z$
g) $\lambda y : \sigma. y$ ✗
h) true false
i) $x \ M$ ✗
j) $\text{if } x \text{ then } \lambda x : \text{Bool}. x$



$(\lambda x: \text{Bool} \rightarrow \text{Bool}. x \ y) (\lambda y: \text{Bool}. x)$



$\Gamma = x: \text{Bool}, y: \text{Bool}$

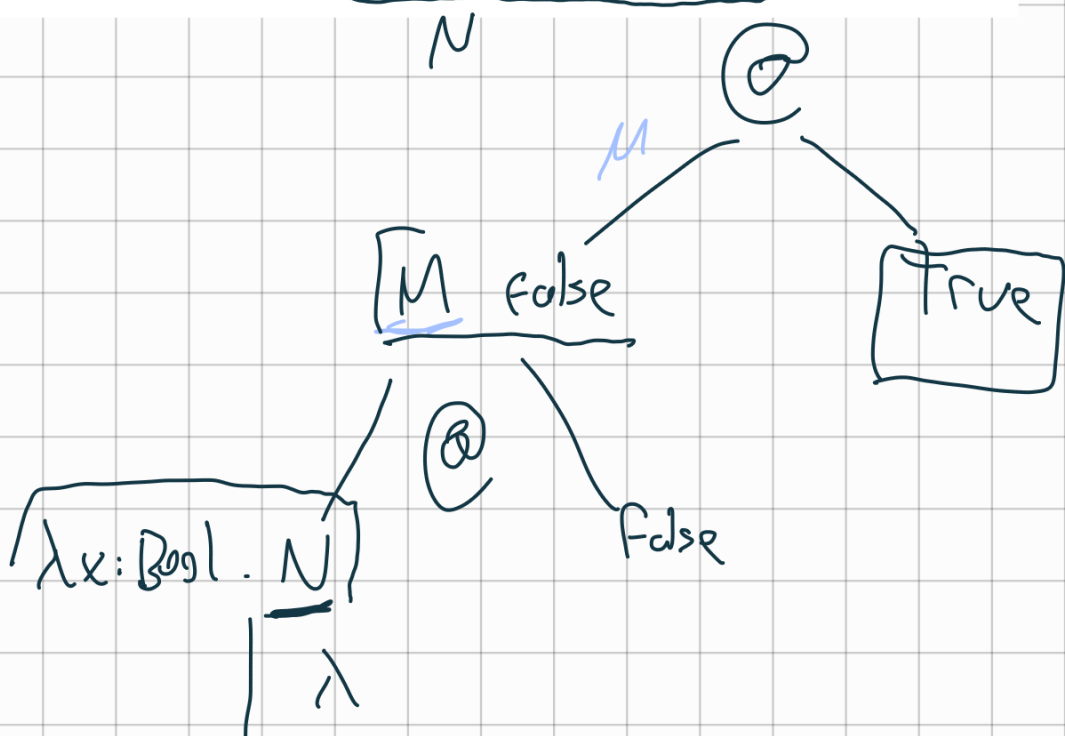
$\frac{}{\Gamma \vdash x: \text{Bool}} \text{T-Var}$	$\frac{}{\Gamma \vdash \text{True}: \text{Bool}} \text{T-True}$	$\frac{}{\Gamma \vdash y: \text{Bool}} \text{T-Var}$	
$x: \text{Bool}, y: \text{Bool} \vdash \text{if } x \text{ then True else } y: \text{Bool}$			
$\frac{}{} \text{T-ABS}$			
$x: \text{Bool} \vdash \lambda y: \text{Bool}. \text{if } x \text{ then True else } y: \text{Bool} \rightarrow \text{Bool}$			
$\frac{}{} \text{T-ABS}$		$\frac{}{} \text{T-false}$	
$\vdash \lambda x: \text{Bool}. \lambda y: \text{Bool}. \text{if } x \text{ then True else } y: \text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}$		$\vdash \text{false}: \text{Bool}$	
$\vdash (\lambda x: \text{Bool}. \lambda y: \text{Bool}. \text{if } x \text{ then true else } y) \text{ false}: \text{Bool} \rightarrow \text{Bool}$			

$$\begin{array}{c}
 \overbrace{\{x : \text{Bool}\} \vdash x : \text{Bool}}^{\text{t-Var}} \quad \overbrace{x : \text{Bool} \vdash x : \text{Bool} \rightarrow \text{Bool}} \\
 \hline
 \{x : \text{Bool}\} \vdash \text{if } x \text{ then } x \text{ else } (\lambda y : \text{Bool}. y) : \text{Bool} \rightarrow \text{Bool} \quad \text{t-if}
 \end{array}$$

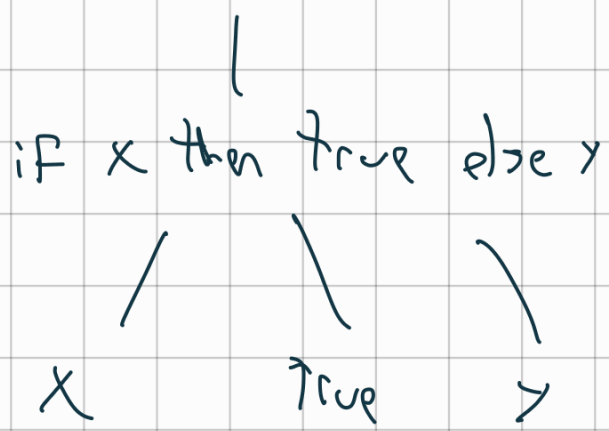
$$\begin{array}{l}
 \rho = \underline{\quad} \rightarrow \underline{\quad} \rightarrow \underline{\quad} \\
 \rho = \underline{\quad} \rightarrow \underline{\quad} \\
 \tau = \underline{\quad} \\
 \sigma = \underline{\quad}
 \end{array}
 \left\{
 \begin{array}{l}
 \rho = \sigma \rightarrow (\tau \rightarrow \underline{\sigma}) \\
 \rho = \underline{\sigma} \rightarrow (\underline{\tau \rightarrow \sigma})
 \end{array}
 \right.$$

$$\begin{array}{c}
 \frac{}{\Gamma \vdash X: \underline{\quad} \rightarrow \underline{\quad} \rightarrow \underline{\quad}} \tau\text{-var} \quad \frac{}{\Gamma \vdash Y: \underline{\quad}} \tau\text{-var} \\
 \frac{}{\Gamma \vdash X: \underline{\quad} \rightarrow \underline{\quad}} \tau\text{-var} \quad \frac{\frac{}{\Gamma \vdash XY: \underline{\quad} \rightarrow \underline{\quad}} \tau\text{-app} \quad \frac{}{\Gamma \vdash Z: \underline{\quad}} \tau\text{-var}}{\Gamma \vdash XYZ: \underline{\quad}} \tau\text{-app} \\
 \frac{}{\Gamma \vdash X: \underline{\quad} \rightarrow \underline{\quad}} \tau\text{-var} \quad \frac{}{\Gamma \vdash XYZ: \underline{\quad}} \tau\text{-app} \\
 \frac{}{\Gamma = x:\rho, y:\sigma, z:\tau \vdash X(XYZ): \underline{\quad}} \tau\text{-app} \\
 \frac{}{x:\rho, y:\sigma \vdash \lambda z:\tau. X(XYZ): \underline{\tau \rightarrow \underline{\quad}}} \tau\text{-Abs} \\
 \frac{}{x:\rho \vdash \lambda y:\sigma. \lambda z:\tau. X(XYZ): \underline{\sigma \rightarrow \underline{\tau \rightarrow \underline{\quad}}}} \tau\text{-Abs} \\
 \frac{}{\vdash \lambda x:\rho. \lambda y:\sigma. \lambda z:\tau. X(XYZ): \rho \rightarrow \underline{\quad}} \tau\text{-Abs}
 \end{array}$$

M
 $((\lambda x : \text{Bool}. \lambda y : \text{Bool}. \text{if } x \text{ then true else } y) \text{ false}) \text{ true}$



$\lambda x : \text{Bool}. \text{if } x \text{ then true else } y$



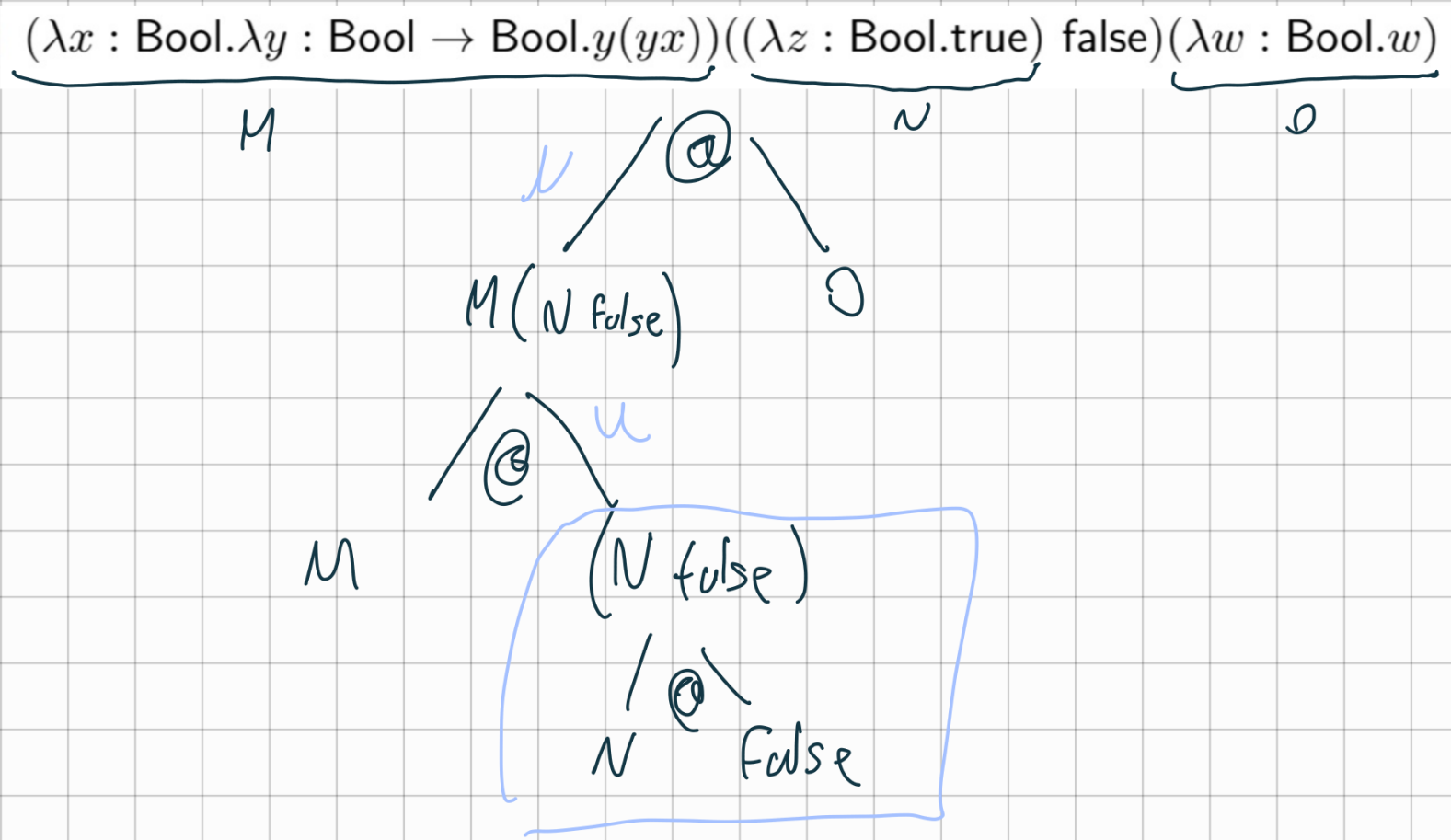
$N \} x := \text{false}$

$((\lambda x : \text{Bool}. \lambda y : \text{Bool}. \text{if } x \text{ then true else } y) \text{ false}) \text{ true}$

$\mu_1 \beta \rightarrow (\lambda y : \text{Bool}. \text{if false then true else } y) \text{ true}$

$\beta \rightarrow \text{if false then true else true}$

$\text{if-f} \rightarrow \text{true}$



$(\lambda x : \text{Bool}. \lambda y : \text{Bool} \rightarrow \text{Bool}. y(yx))((\lambda z : \text{Bool}. \text{true}) \text{ false})(\lambda w : \text{Bool}. w)$

$$\xrightarrow{\nu, u, \beta} (M \text{ true}) O$$

$$\xrightarrow{\nu, \beta} (\lambda y : \text{Bool} \rightarrow \text{Bool}. y(y \text{ true})) (\lambda w : \text{Bool}. w)$$

$$\xrightarrow{\beta} (\lambda w : \text{Bool}. w) ((\lambda w : \text{Bool}. w) \text{ true})$$

$$\xrightarrow{u, \beta} (\lambda w : \text{Bool}. w) \text{ true}$$

$$\xrightarrow{\beta} \text{true}$$

$$\begin{array}{c}
\frac{}{x:\text{Nat} \vdash x:\text{Nat}} \text{T-Var} \\
\frac{}{x:\text{Nat} \vdash \text{Succ}(x):\text{Nat}} \text{T-Succ} \\
\frac{}{\vdash \lambda x:\text{Nat}. \text{Succ}(x) : \text{Nat} \rightarrow \text{Nat}} \text{T-Abs} \quad \frac{}{\vdash \text{Zero}:\text{Nat}} \text{T-Zero} \\
\hline
\vdash (\lambda x:\text{Nat}. \text{Succ}(x)) \text{Zero} : \text{Nat} \quad \text{T-APP}
\end{array}$$

Zero

isZero(succ(pred(succ(Zero))))

$$\begin{array}{l}
\longrightarrow \text{isZero}(\text{succ}(\text{Zero})) \longrightarrow \text{False} \\
\text{E-isZero} \quad \text{E-isZero}_n
\end{array}$$

E-Succ
E-predSucc

isZero(succ(pred(Zero)))

$$\begin{array}{l}
\longrightarrow \\
\text{E-isZero}
\end{array}$$

Caso Base

$$- M \rightarrow_{\beta} M_1, M \rightarrow_c M_2 \quad \text{gracias} \quad M_2 = M_1$$

$$M = (\lambda x: \sigma. N) V$$

$$- M \rightarrow_{\text{ift}} M_1, M \rightarrow_{\text{let}} M_2 \quad \text{gracias} \quad M_2 = M_1$$

$$- \text{Lo mismo con } M \rightarrow_{\text{ifF}} M_1$$

Caso inductivo:

$$\text{HI: } M' \rightarrow M'_1, M' \rightarrow M'_2 \quad \Rightarrow \quad \overline{M'_1 = M'_2}$$

Caso μ

$$M \rightarrow_{\mu} M_1 \quad \text{p.e.} \quad M = M' N, \quad M_1 = M'_1 N$$

$$\text{donde } M' \rightarrow M'_1$$

$$M \rightarrow M_2$$

$$M'N \rightarrow M'_2 N \quad \text{si} \quad M' \rightarrow M'_2$$

$$M' \rightarrow M'_1 \quad , \quad M' \rightarrow M'_2 \quad \Rightarrow \quad \underline{M'_1} = M'_2$$

$$\underbrace{M'_2 N} = \underbrace{M'_1 N}$$

$$M_2 = M_1$$

$$(\lambda x: \text{Bool} \rightarrow \text{Bool}. X) \text{ True}$$